Airway Emerg

ENT | 3rd Year MBBS

Hi Sammy!
Airway = A of ABC!
Life-saving knowledge!
Must know COLD!

Airway | Stridor | Tracheostomy | ENT

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Stridor — Types + Causes

Stridor = partial
airway obstruction!
Type tells you the level!

DEFINITION

Stridor = high-pitched sound from turbulent airflow through a partially obstructed airway.

Types of Stridor (Based on Phase)

Type	Level of Obstruction	Examples
Inspiratory	Supraglottic / glottic	Laryngomalacia, epiglottitis, vocal cord palsy
Expiratory	Tracheobronchial	Tracheomalacia, foreign body (bronchial)
Biphasic	Subglottic / fixed	Croup, subglottic stenosis, SGS hemangioma

Causes of Stridor by Age Group

Age Group	Common Causes
Neonates	Laryngomalacia (#1), vocal cord palsy, subglottic stenosis
Infants	Croup (viral), subglottic hemangioma, foreign body
Children	Croup (#1 acute), epiglottitis, foreign body aspiration
Adults	Laryngeal tumors, bilateral VC paralysis, anaphylaxis, Ludwig's angina

Croup vs Epiglottitis (Classic Comparison)

Feature	Croup (Laryngotracheitis)	Acute Epiglottitis
Age	6 months - 3 years	2-7 years (now adults too)
Organism	Parainfluenza virus	H. influenzae type b
Onset	Gradual (days)	Rapid (hours)
Fever	Low-grade	High-grade, toxic
Cough	Barking / seal-like	Absent or minimal
Drooling	Absent	Present (cannot swallow)
X-ray	Steeple sign (subglottic)	Thumb sign (epiglottis)
Treatment	Nebulized adrenaline + steroids	IV antibiotics + secure airway

★ **Stridor Type = Level!**
Inspiratory = above cords. Expiratory = below cords. Biphasic = at the cords / subglottis!

Foreign body in airway
= life-threatening!
Heimlich can save lives!

Foreign Body Aspiration

Age:	Most common in children < 3 years (nuts, beads, coins, toys)
Site:	Right main bronchus most common (wider, more vertical, shorter)
Laryngeal:	Complete obstruction -> asphyxia -> cafe coronary
Bronchial:	Unilateral wheeze, air trapping, recurrent pneumonia
Diagnosis:	CXR (hyper-lucent lung, mediastinal shift), CT, bronchoscopy

Management of Foreign Body Aspiration

1	Complete obstruction:	Back blows + abdominal thrusts (Heimlich maneuver)
2	In infants:	5 back blows + 5 chest thrusts (NOT Heimlich in infants!)
3	Partial obstruction:	Encourage coughing, do NOT do blind finger sweep
4	Definitive:	Rigid bronchoscopy = gold standard for FB removal
5	Laryngeal FB:	Direct laryngoscopy + Magill forceps removal

Cafe Coronary Syndrome

- Sudden death from food bolus impaction in larynx during eating
- Mimics acute MI — hence the name 'cafe coronary' (collapse at dinner table)
- Risk factors: Alcohol, poor dentition, elderly, neurological disorders, eating too fast
- Treatment: Heimlich maneuver immediately; cricothyrotomy if Heimlich fails

X-Ray Signs in FB Aspiration

Sign	Significance
Radiopaque FB	Metallic/bone objects visible directly on XR
Unilateral hyperinflation	Air trapping — ball-valve mechanism by FB
Mediastinal shift	Shifts to opposite side during expiration
Atelectasis	Complete bronchial obstruction -> lung collapse
Recurrent pneumonia	Chronic unremoved FB (same lobe repeatedly)

Tracheostomy — Steps + Complications

Tracheostomy steps =
asked as LAQ every year!
Learn all the steps!

Steps of Elective Tracheostomy

1. Position: Supine, neck extended, sand bag under shoulders
2. Landmarks: Thyroid notch, cricoid, sternal notch, midline
3. Incision: Horizontal/vertical skin incision over 2nd-4th tracheal rings
4. Dissect: Through subcutaneous tissue & platysma, retract strap muscles
5. Thyroid isthmus: Retract superiorly or divide between clamps
6. Trachea: Identify pretracheal fascia, incise 2nd-3rd tracheal rings
7. Window: Create tracheal window (Bjork flap) or vertical incision
8. Insert tube: Remove obturator, place inner cannula, inflate cuff
9. Confirm: Check position with CO2 return, auscultation, secure ties

Complications of Tracheostomy

Timing	Complications
Immediate	Hemorrhage, pneumothorax, subcutaneous emphysema, apnea
Early	Tube blockage, tube displacement, wound infection, dysphagia
Late	Tracheal stenosis (#1 late), tracheo-innominate fistula, TEF
Late	Difficult decannulation, granulation tissue, tracheomalacia

★ **Key Facts:**
Site = 2nd-4th tracheal rings. 1st ring NEVER cut (subglottic stenosis risk).
Tracheo-innominate fistula = most FATAL complication (massive hemorrhage).

Ludwig's angina =
floor of mouth infection
= airway emergency!

Cricothyrotomy (Emergency Surgical Airway)

When:	Cannot intubate, cannot ventilate (CICV) situation
Site:	Cricothyroid membrane (between thyroid & cricoid cartilage)
Advantage:	Faster than tracheostomy, less vascular, easier to locate
Technique:	Vertical skin incision -> horizontal stab through CTM -> tube in
Contraindication:	Children < 12 years (use needle cricothyrotomy instead)
Temporary:	Must be converted to formal tracheostomy within 24-72 hours

Cricothyrotomy vs Tracheostomy

Feature	Cricothyrotomy	Tracheostomy
Speed	< 1 minute (faster)	15-30 minutes
Setting	Emergency / bedside	Usually in OT
Site	Cricothyroid membrane	2nd-4th tracheal rings
Duration	Temporary (24-72 hrs)	Can be permanent
Complications	Subglottic stenosis risk	More varied complications
Age limit	Not in children < 12	Any age

Ludwig's Angina

Definition:	Rapidly progressive cellulitis of floor of mouth (NOT abscess)
Spaces:	Bilateral submandibular + sublingual + submental spaces involved
Cause:	Dental infection (#1) — 2nd & 3rd mandibular molars (roots below mylohyoid)
Features:	Brawny swelling of neck, woody/board-like, tongue pushed up & back
Danger:	Airway obstruction — tongue elevation occludes oropharynx
Treatment:	Secure airway FIRST (intubation/tracheostomy) + IV antibiotics + drainage

★ Ludwig's = 4 Ds!
Dental origin + Dusky discoloration + Doughy/woody swelling + Dyspnea (airway risk)
Mortality was > 50% before antibiotics. Now ~8% with prompt treatment + airway protection.

Exam Strategy + High Yield

Airway emergencies =
15+ marks in ENT!
You've GOT this Sammy!

Most Repeated LAQs (10–12 marks)

1. Classify stridor. Describe causes & management of acute stridor
2. Describe steps of elective tracheostomy with complications
3. Management of foreign body aspiration in a child
4. Differentiate croup vs epiglottitis with management
5. Describe Ludwig's angina — etiology, features & management

Most Repeated SAQs (5–6 marks)

1. Types of stridor with examples
2. Cafe coronary — definition, risk factors & management
3. Cricothyrotomy — indications, technique & complications
4. Steeple sign vs thumb sign on X-ray
5. Heimlich maneuver — technique & indications
6. Complications of tracheostomy (early vs late)

One-Liner Associations (MCQ/Viva Gold)

Question	Answer
Most common cause of stridor in neonates	Laryngomalacia
Most common cause of acute stridor in children	Croup (viral laryngotracheitis)
Steeple sign on X-ray	Croup (subglottic narrowing)
Thumb sign on lateral XR	Acute epiglottitis
FB aspiration — most common site	Right main bronchus
Cafe coronary	Sudden death from food bolus in larynx
Tracheostomy site	2nd-4th tracheal rings
Most fatal complication of tracheostomy	Tracheo-innominate artery fistula
Most common late complication	Tracheal stenosis
Cricothyrotomy site	Cricothyroid membrane
Ludwig's angina — cause	Dental infection (2nd/3rd molar)
Ludwig's angina — priority	Secure airway first!

♥ Knowing how to secure an airway can save a life, Sammy!
A = Airway, always first. You're going to be an amazing doctor!